# Machine Learning-Driven Dollar Neutral Equity Strategy

## Volatility Targeting & Monte Carlo Significance Testing

Out-of-Sample Evidence from 1957–1990 | Rolling OOS + Monte Carlo Validation

OUR TEAM

# Meet the Team

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EXECUTIVE SUMMARY

# What Did We Find?

1

### Nonlinear ML Outperforms

A **dollar-neutral long–short strategy** delivers economically meaningful **out-of-sample performance** when powered by **nonlinear machine learning**.

2

### LightGBM Model Results

Using a **10% long/10% short portfolio**, the **LightGBM model** achieves:

- **Sharpe = 0.82** (raw strategy)
- **Sharpe = 0.69** (with volatility targeting)
  This **substantially outperforms** the linear **Elastic Net benchmark** (Sharpe ≈ 0.01).

3

### Volatility Targeting Benefits

**Volatility targeting** stabilizes risk over time while preserving return potential, resulting in comparable drawdowns and improved risk consistency.

4

### Overall Conclusion

Nonlinear ML improves **cross-sectional stock ranking**, and volatility targeting provides **effective risk control without eliminating alpha**.

| Model | Vol Targeted | Ann. Return | Ann. Vol | Sharpe | Max Drawdown |
|---|---|---|---|---|---|
| LightGBM | **No** | 12.28% | 15.47% | **0.8228** | -18.73% |
| LightGBM | Yes | 9.70% | 14.83% | **0.6936** | -18.73% |
| ElasticNet | No | -0.27% | 8.72% | 0.0134 | -20.30% |

VALIDATION DESIGN

# Why We Trust the Results: Rolling Out-of-Sample

Every return in the equity curve is generated by a model that had **never seen** that month's data during training or tuning.

- **No Look-Ahead Bias**
  Training data strictly precedes evaluation periods.
- **Reduced Overfitting Claims**
  Rolling retraining enforces generalization across time and market regimes.
- **Replicability**
  Window lengths and step sizes are fixed in code and consistently applied.

PORTFOLIO CONSTRUCTION

# Translating Predictions into Tradable Alpha

Go long the top 10% and short the bottom 10%, equally weighted within each leg. Dollar-neutral structure (long +1, short -1) isolates the signal's cross-sectional predictive power. Monthly return: R_t = R_long - R_short.

**Each month, rank stocks by predicted excess return.**

1

### Rank All Stocks

Rank investable universe by out-of-sample predicted excess return.

2

### Select Extremes

Each month, stocks are ranked by predicted excess return and transformed into a long–short portfolio. Long top 10% | Short bottom 10% | Middle ranks receive zero weight.

3

### Equal Weight Within Legs

Equal-weight positions to isolate ranking signal from sizing effects.

4

### Dollar-Neutral by Construction

Long and short notional matched monthly to eliminate market beta.

VOLATILITY TARGETTING

# Why Volatility Targeting Improves Comparability

Volatility targeting addresses time-varying volatility in raw long-short returns by dynamically scaling positions to a common risk level, preventing distortion of aggregate risk statistics and Sharpe ratios.

1. **Estimate Volatility**

   Estimates trailing 36-month rolling standard deviation of annualized returns to capture prevailing risk.

2. **Scale to Target**

   Scales raw returns by realized volatility, multiplying by a **10% target** to achieve comparable risk levels.

3. **Lag by One Month**

   Lags volatility estimate by one month (t-1) to avoid look-ahead bias and ensure observability.

4. **Cap Leverage**

   Applies a leverage cap to prevent excessive scaling; in our sample, it's **never binding**, ensuring stable risk.

*Volatility targeting is applied at the portfolio level after long–short aggregation.*

**Implementation Diagnostics:** Avg. long positions/month: ~486 • Avg. short positions/month: ~486 • Avg. gross leverage: ~0.84 • Leverage cap binding: 0% of months • Avg. monthly turnover: ~0.92 *(Computed directly from portfolio construction step.)*

EQUITY CURVE

# Equity Curve: ML vs Linear Benchmark

## LightGBM

- Starts at $1
- Grows to ~2.3–2.5
- Shows sustained upward trajectory

## ElasticNET

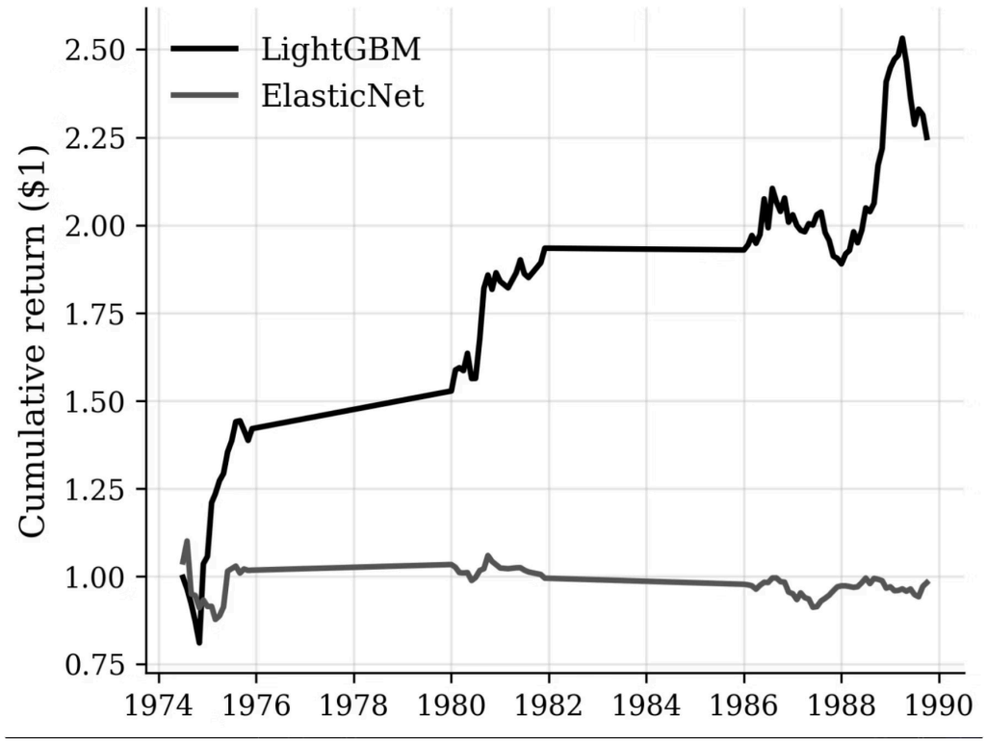
- Stays around $1
- Slight drift downward over time
- No meaningful wealth creation

## Results

- **LightGBM** strategy delivers sustained positive cumulative returns
- **Elastic Net** benchmark remains near flat
- Performance divergence highlights the value of **nonlinear ranking**

## Interpretation

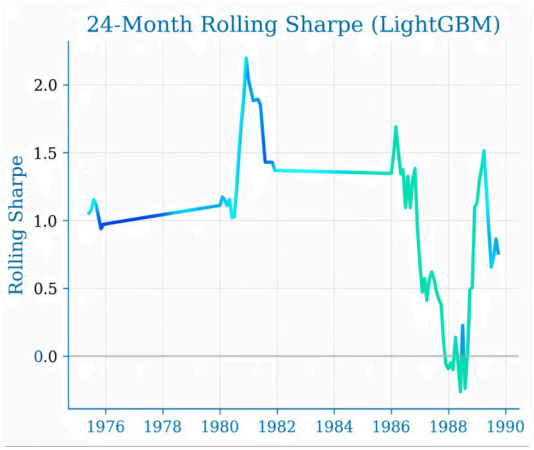
- Returns are driven by **relative ordering**, not point forecasts
- Confirms ML improves **cross-sectional stock selection**, not market timing

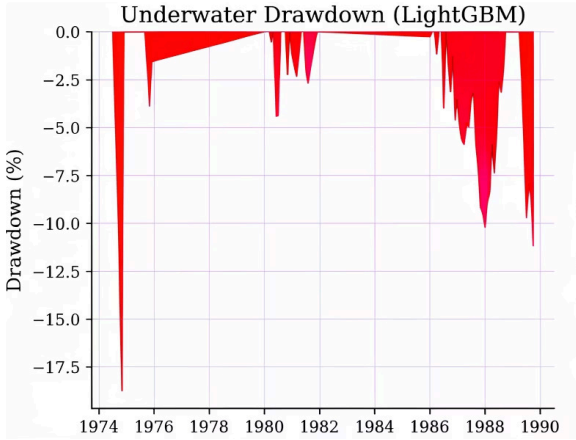


RISK PROFILE

# Is the Performance Stable

*Performance is **variable but resilient**:.*




## Rolling Sharpe Ratio

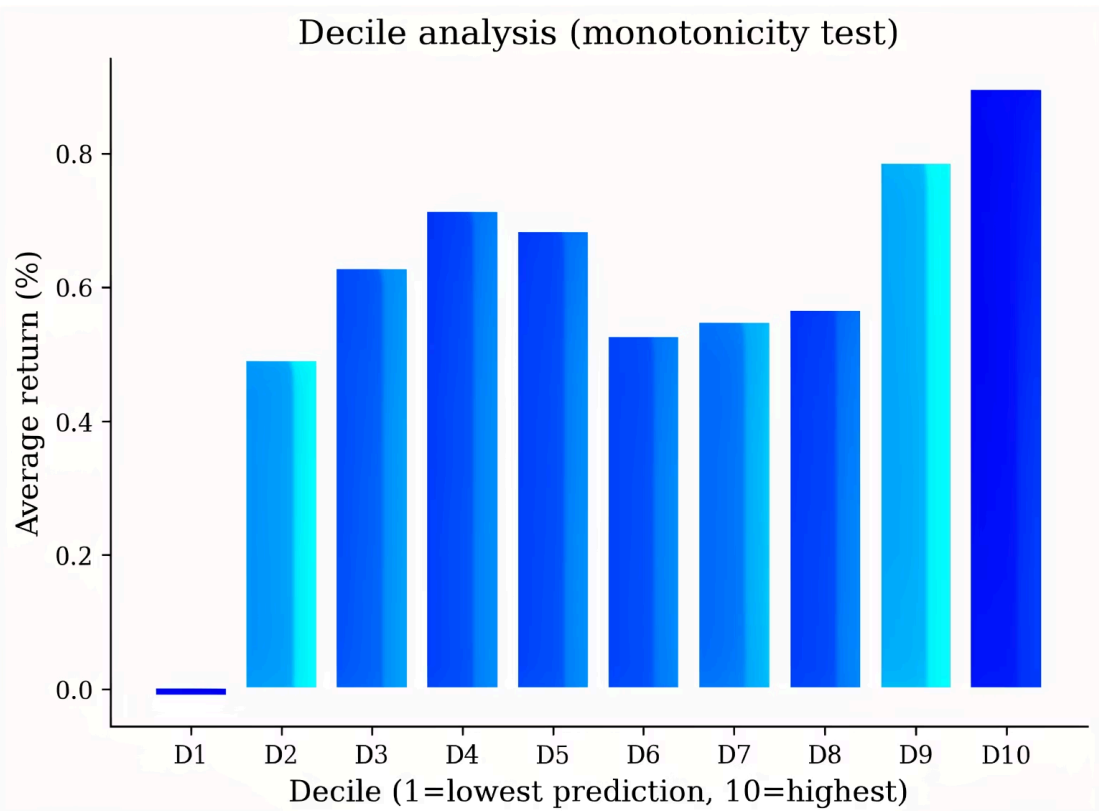
- The 24-month rolling Sharpe varies across market regimes
- Periods of weaker performance are followed by recoveries
- Risk-adjusted returns are **not driven by a single favorable window**

## Drawdown Behavior

- Drawdowns occur, including several short-lived deeper episodes
- Losses remain **controlled relative to cumulative gains**
- The strategy consistently **recovers after drawdowns**

DECILE MONOTONICITY TEST

# Decile Monotonicity Spread Confirms Signal Quality




Deciles formed cross-sectionally each month using model scores.

### Decile Monotonicity Confirms Ranking Power

- Stocks are sorted monthly by predicted scores.
- Average returns rise from bottom to top decile.
- The monotonic spread validates cross-sectional predictive strength.

### Results

- Returns increase approximately monotonically from the lowest to highest decile.
- This indicates the model provides meaningful cross-sectional ranking information.

### Implication

The signal is valuable for relative ordering, supporting long–short portfolios constructed from extreme ranks rather than relying on precise point forecasts.

INFORMATION COEFFICIENT

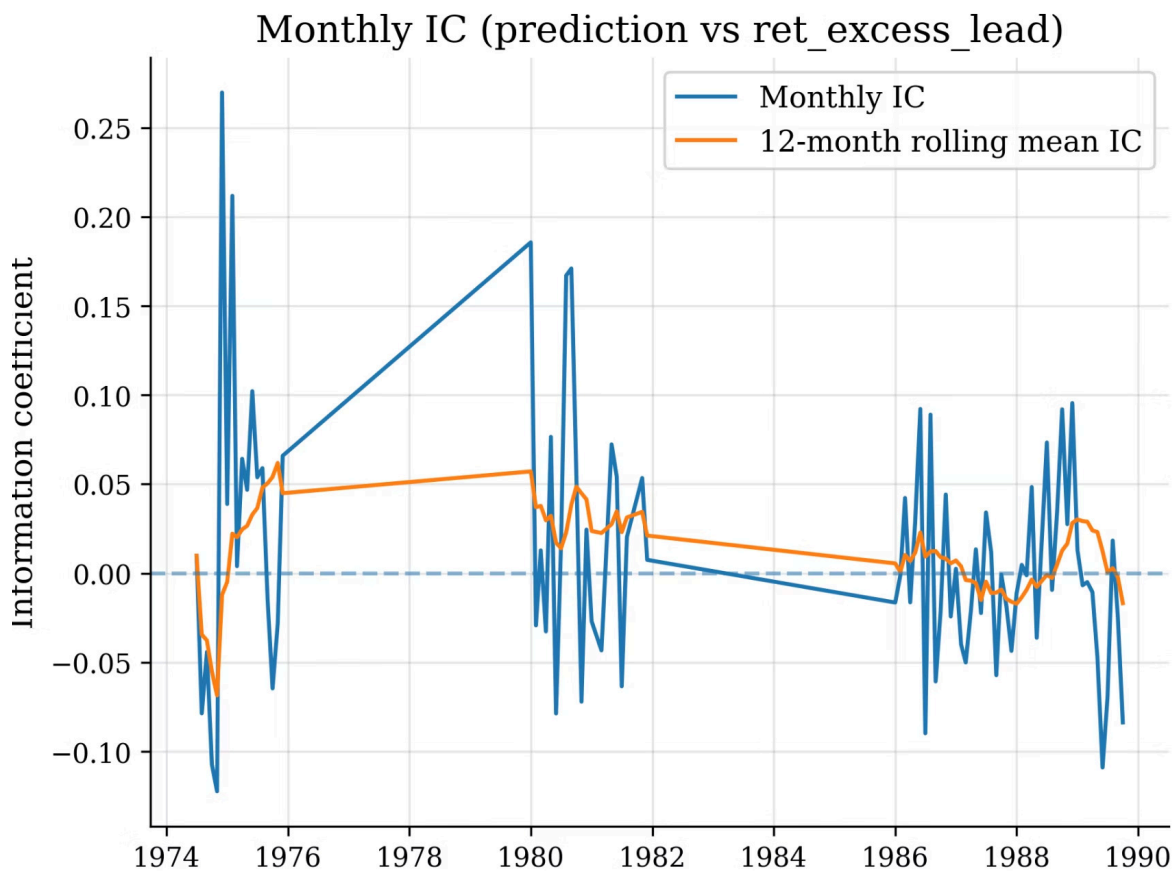
# Does the Signal behave like Alpha?

### Information Coefficient (IC) Over Time

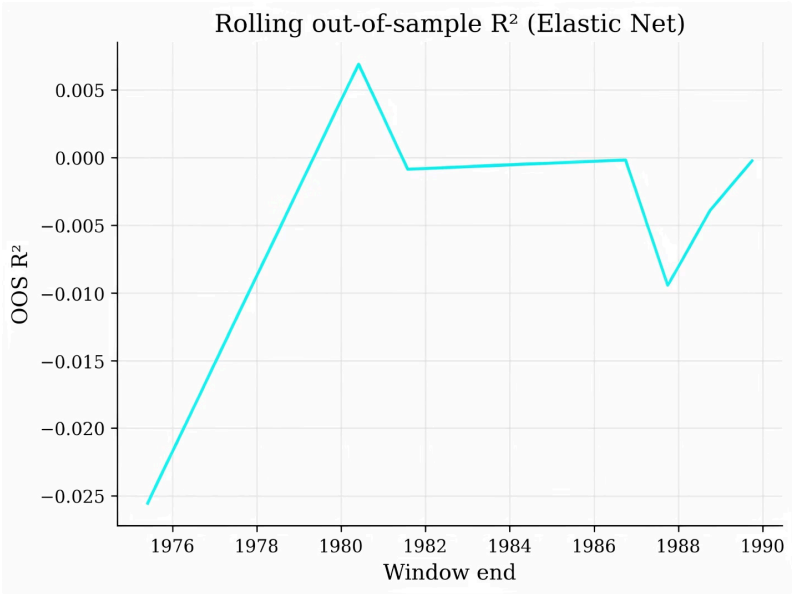
IC measures the rank correlation between model scores and next-month excess returns

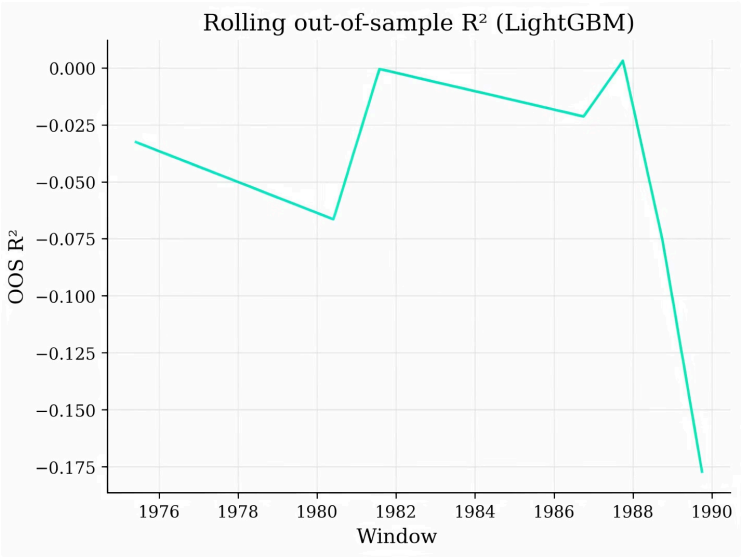
- Monthly IC is noisy due to idiosyncratic stock-level variation
- The 12-month rolling IC remains modestly positive over most of the sample
- Even small positive IC values are economically meaningful in large cross-sections
- Persistent positive IC indicates stable cross-sectional ranking skill, not point forecasting

The model does not predict exact returns, but it consistently ranks stocks correctly, which is the core requirement for cross-sectional alpha.



## Why $R^2$ Is Low and Why That's Expected




**1** Rolling out-of-sample $R^2$ evaluates point-return prediction accuracy, not cross-sectional ranking skill.

**2** Monthly individual stock returns are dominated by idiosyncratic noise, so out-of-sample $R^2$ is typically near zero or negative.

**3** Both ElasticNet and LightGBBM exhibit low OOS $R^2$, which is consistent with asset pricing and empirical finance literature.

**4** Our strategy does not rely on point forecasts; performance is driven by relative ranking, evaluated using IC, decile monotonicity, and long–short returns.

MONTE CARLO VALIDATION

# Monte Carlo Significance Testing

**Is the Alpha Statistically Real?**

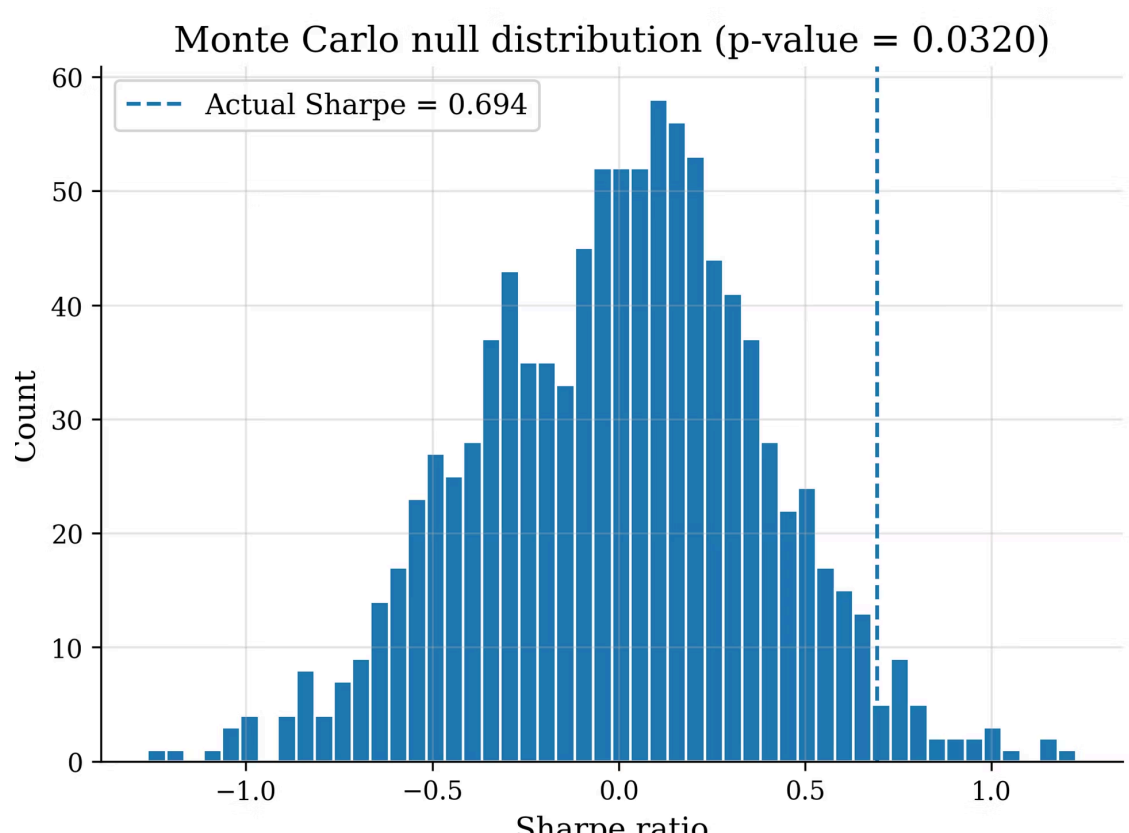
We apply a **Monte Carlo based robustness check** using randomized simulations of the strategy.

### Methodology

- Generate randomized return paths consistent with the strategy's structure
- Compute Sharpe ratios across thousands of simulated trials
- Construct an empirical **null distribution of Sharpe outcomes**

### Results

- The actual strategy Sharpe (0.69) lies in the **extreme upper tail** of the simulated distribution
- Estimated p-value ≈ 0.032
- Fewer than 3.2% of random simulations achieve comparable performance




Monte Carlo analysis is used as a **robustness stress test**, complementing rolling out-of-sample validation and signal diagnostics.

⚠ CHALLENGES

# Challenges We Had to Handle

Building a credible ML alpha strategy over six decades of U.S. equity data surfaced several non-trivial obstacles — each requiring deliberate engineering decisions, not textbook defaults.

### Extremely Low Signal-to-Noise

Monthly individual stock returns are dominated by idiosyncratic noise. Even a well-specified model yields IC values in the 0.02–0.05 range. Aggregation across hundreds of stocks is essential to extract a meaningful portfolio-level signal.

### Overfitting Risk in High Dimensions

With dozens of features and a flexible tree ensemble, the model can memorize in-sample patterns. We mitigate this risk through rolling retraining, regularization, and strict out-of-sample evaluation

### Missing Data & Preprocessing Sensitivity

The 1957–2021 panel has significant missing data in early decades. Imputation, forward-filling, and feature availability cutoffs all influence results. We document every choice.

### Feature Drift Across Decades

Factor premia shift over time. A feature predictive in the 1970s may lose power by the 2010s. Rolling retraining partially addresses this, but structural breaks remain a concern.

# Thank you - Questions?

We welcome your questions.